

Qualitative Phytochemical Analysis

Qualitative phytochemical screening was performed to identify the presence of bioactive compounds.

Test for Alkaloids

Dried extract (0.1 g) was dissolved in ethanol and filtered. The filtrate was acidified using hydrochloric acid and treated with Wagner's reagent. The formation of a reddish-brown precipitate indicated alkaloids.

Test for Saponins

Extract was mixed with distilled water and vortexed. The formation of stable foam indicated saponins.

Test for Flavonoids

Ethanol extract was treated with sodium chloride and hydrochloric acid. A color change indicated flavonoids.

Test for Tannins

Aqueous extract was treated with ferric chloride. A blue-black or greenish coloration indicated tannins.

Test for Terpenoids (Salkowski Test)

Extract was mixed with chloroform and concentrated sulfuric acid. A reddish-brown interface indicated terpenoids.

Test for Quinones

Extract was treated with sodium hydroxide solution. Formation of a colored solution indicated quinones.

Test for Cardiac Glycosides (Keller–Killiani Test)

Extract was treated with chloroform, followed by acetic acid containing ferric chloride and concentrated sulfuric acid. Formation of a brown ring indicated cardiac glycosides.

GC–MS Analysis

GC–MS analysis was performed to identify phytochemical constituents present in the *Acalypha indica* extract. The analysis was carried out using a gas chromatograph coupled with a mass spectrometer equipped with a DB-5 capillary column. The oven temperature was programmed from 100 °C to 270 °C at a rate of 10 °C/min. Helium was used as the carrier gas at a constant flow rate of 1.2 mL/min. The mass spectrometer

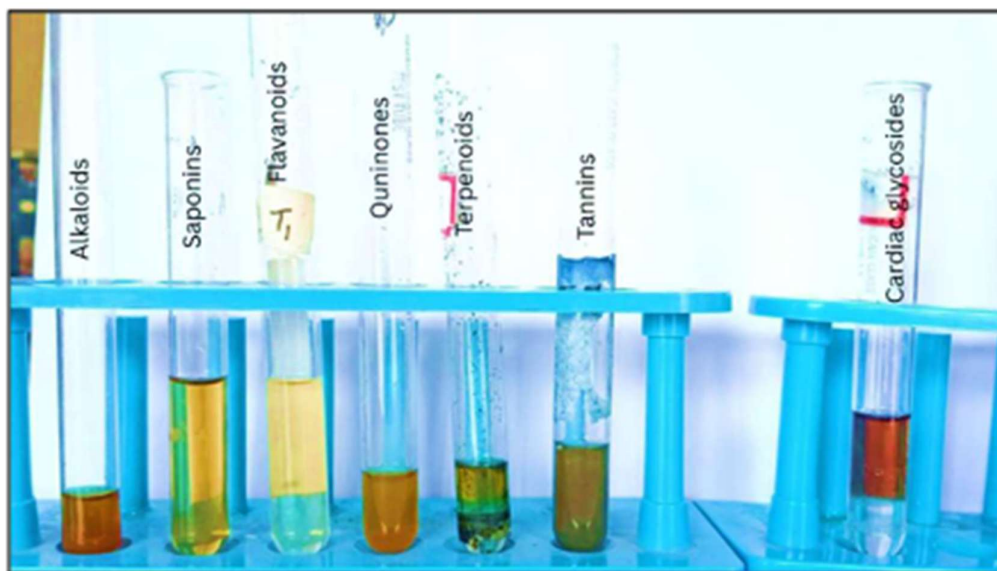
operated in scan mode. Compounds were identified based on their retention time and comparison with standard mass spectral libraries

RESULTS

PHYTOCHEMICAL ANALYSIS

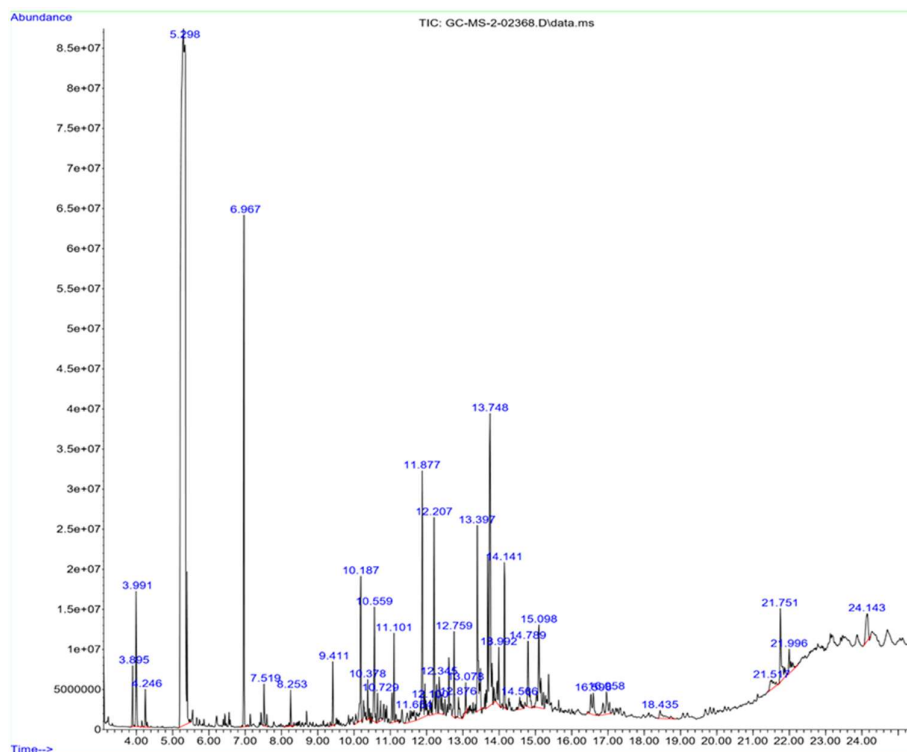
S.No	Phytochemical	Results
1	Alkaloids	+++
2	Saponins	–
3	Flavonoids	+++
4	Tannins	+++
5	Terpenoids	++
6	Quinones	+++
7	Cardiac glycosides	++

– Not detected | + Mild positive| ++ Moderate positive| +++ Strong positive



*Saturation of the image has been increased to enhance the clarity of the image

GAS CHROMATOGRAPHY-MASS SPECTROMETRY ANALYSIS



Major compounds identified

S.no	Retention Time (min)	Compound Name	Molecular Formula	Molecular Weight	Peak Area (%)
1	5.298	Phenol	C ₆ H ₆ O	94.11	50.19
2	6.967	Acetic acid, 2-ethylhexyl ester	C ₁₀ H ₂₀ O ₂	172.26	5.22
3	10.378	2,4-di-tert-butylphenol	C ₁₄ H ₂₂ O	206.32	0.49
4	11.877	Hexadecane, 2,6,10,14-tetramethyl-	C ₂₀ H ₄₂	282.55	3.98
5	12.100	Pyridine-3-carboxamide, oxime	C ₆ H ₇ N ₃ O	137.14	0.31
6	12.207	Hexadecane, 2-methyl-	C ₁₇ H ₃₆	240.47	2.77
7	12.759	Tetradecanal	C ₁₄ H ₂₈ O	212.37	1.00
8	13.078	1,2-Benzenedicarboxylic acid	C ₈ H ₆ O ₄	166.13	0.34

Among these, 2,4-di-tert-butylphenol is reported to exhibit antioxidant and cytotoxic activities, making it a promising candidate for anticancer evaluation